

Question 1

Aim

To solve the recurrence relation $T(n) = 2T(n/2) + n$ using the Master Theorem and determine its time complexity.

Procedure

1. Start the program.
2. Read the number of users (n) from the user.
3. Identify the recurrence relation $T(n) = 2T(n/2) + n$.
4. Apply the Master Theorem.
5. Find the values of a , b , and $f(n)$.
6. Compare $f(n)$ with $n^{\log_2 a}$.
7. Identify the correct Master Theorem case.
8. Calculate the time complexity.
9. Display the approximate operations.
10. Stop the program.

Algorithm

1. Start.
2. Input the value of n .
3. Set $a = 2$, $b = 2$, and $f(n) = n$.
4. Calculate $n \times \log_2(n)$.
5. Display the approximate operations.
6. Display the time complexity $\Theta(n \log n)$.
7. Stop.

Input

Enter number of users: 1024

Output

Approximate operations = 10240.0

Time Complexity = $\Theta(n \log n)$

Result

The recurrence relation $T(n) = 2T(n/2) + n$ was solved using the Master Theorem. The algorithm satisfies Case 2, and the final time complexity is $\Theta(n \log n)$.